

Topic: Projection Operation

Q. Discuss the Projection operation in Relational Algebra. Describe its syntax and explain its behavior regarding duplicate elimination, using suitable examples.

[DEF] Definition to Remember

The Projection Operation (denoted by π) is a unary operation in relational algebra that filters specific columns (attributes) from a relation while discarding the rest. It performs vertical filtering and automatically eliminates duplicate rows in the result. It is equivalent to the `SELECT column_name` in SQL.

Projection Operation (π)

* Operates on a single table (unary).

* Fil

2. Syntax of Projection

$\pi_{\{<attribute_list>\}}(Relation_Name)$

* attribute_list: Comma-separated column names to retain.

3. Examples

Input Relation: STUDENT

Roll_No	Name	Major	City
101	John	CS	New York
102	Emma	Math	Boston
103	Luke	CS	New York
104	Mia	Physics	Chicago

Example 1: Simple Projection

Query:

Example 2: Duplicate Elimination

Query:

Example 3: Composition -- Selection + Projection

Query:

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Projection is a unary operation -- operates on one relation.
2. Denoted by π (Π); equivalent to SQL's `SELECT column_name`.
3. Performs vertical filtering -- selects specific columns, discards others.
4. Critical feature: automatically eliminates duplicate rows from the result.
5. The number of columns in output = number of columns listed in the attribute list.

6. Can be composed with Selection -- Selection applied first, Projection second.

7. Duplicate elimination is automatic because a relation is a mathematical set.

[RECALL] Quick Recall

π (Π) -> Unary -> Vertical Filtering (columns) -> Duplicate Elimination (set) -> Equivalent to SQL SELECT -> Composable with σ